When Structure Is the Model: Exact Epistemic Coordinates, Cross-Domain Transfer, and the Limits of Neural Escalation

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Abstract

Learning-system failures can arise because a representation omits information, because a learner fails to exploit information that is present, or because the remaining operation is computational rather than statistical. We study these cases in a prospectively frozen hierarchy of exact synthetic and procedural method-structure tasks. Model-visible views are restricted so that observational collisions produce exact empirical deterministic ceilings on the frozen sample. Generic classical learners receive first right of refusal; when an information-sufficient view still leaves a residual, a separately frozen explicit-inference control tests whether computation closes it before neural complexity is escalated. In M1, the semantic view supports mechanic selection at 1.000000, while affine gluing remains at 0.510417despite an exact task ceiling of 1.000000; exact affine composition closes that residual. On a whole held-out domain of transactional workflows, typed relational method comparisons reach 1.000000protected accuracy, versus 0.906250for untyped pair structure, 0.500000for a same-information typed serialization, and 0.250000for reminted transcript features, which is the base rate of the majority class: the transcript view mints a fresh symbol per instance, so its vocabulary does not transfer at all and its classifier is constant. We do not introduce a new graph, sheaf, neuro-symbolic, or latent architecture. The contribution is a bounded diagnostic methodology and evidence object for separating information, learning, computation, and transfer before architecture escalation.

1 Introduction

A poor benchmark score does not uniquely diagnose a poor model. Two task instances may be indistinguishable under the representation given to the learner, in which case additional capacity cannot recover information that was never exposed. Conversely, a representation may contain enough information while a selected model family still fails because the target requires composition, search, or another operation that the predictor does not execute. Treating both situations as invitations to add parameters or test-time computation confounds representation quality with learning and inference.

This paper asks a deliberately bounded question: *what can be learned, transferred, or inferred once representation identifiability is controlled explicitly?* We construct exact hostile pairs, restrict what a learner can observe, and derive the best deterministic accuracy possible on the frozen sample under each view. We then separate four quantities:

1. **information sufficiency**: whether a model-visible view can distinguish the protected targets;
2. **learning sufficiency**: whether a generic learner reaches the performance permitted by an information-sufficient view;
3. **computational sufficiency**: whether a separately frozen explicit procedure closes a residual once the required coordinates are visible;
4. **transfer**: whether a representation learned or scored on some procedural domains remains useful on a whole held-out domain with reminted surface identities and protected corruptions.

The broader ingredients are not new. Relational and graph inductive biases are established in graph Transformers and heterogeneous attention (Ying et al., 2021; Hu et al., 2020; Rampášek et al., 2022); local vector spaces and relation maps are established in sheaf-based representation learning (Gebhart et al., 2021; Bodnar et al., 2022); neural algorithmic reasoning and Transformer–algorithmic-reasoner hybrids already test structured execution and out-of-distribution generalization (Veličković et al., 2022; Bounsi et al., 2024). Reusable abstractions and libraries are central to DreamCoder and LILO (Ellis et al., 2020; Grand et al., 2023); reasoning outside token space is studied by continuous-latent methods such as Coconut (Hao et al., 2024); rule/entity factorization is established by Neural Production Systems (Goyal et al., 2021); and mechanism-centric scientific representations are an active research direction (Posner et al., 2026).

The diagnostic principle is also not ours in the abstract. Recent work explicitly separates representation or perception failures from downstream reasoning failures (Liem, 2026), and deterministic benchmarks with symbolic–neural oracle controls can isolate algorithmic choke points (Anonymous, 2026). Missing-information benchmarks likewise ask systems to recognize when a task is underspecified and information must be acquired rather than guessed (Li et al., 2025). We therefore make no claim to be the first to separate representation from reasoning failure, and the collision-derived ceiling below is not a new theorem. Our residual contribution is the combined controlled study: exact view collisions, simple-learning first right of refusal, explicit computational closure, and protected cross-domain method-coordinate transfer under weaker and same-information controls.

The study supports three bounded conclusions. First, specified typed coordinates are genuinely load-bearing on exact hostile pairs: weaker views are non-identifying by construction rather than merely difficult. Second, the remaining M1 affine-gluing gap is computational: the semantic view contains enough information, but a frozen generic classical grid stays near chance while an exact payload-only composition closes the operation. Third, typed relational method comparisons transfer on a protected whole-domain holdout beyond untyped structure and a same-information typed serialization, both of which respond to the holdout; the reminted transcript view does not transfer at all, and its gap is the majority-class base rate rather than a measured margin. No neural architecture is required by the final bounded evidence.

2 Related work and claim subtraction

Structured representations. Graphormer, Heterogeneous Graph Transformer, and GraphGPS establish multiple ways to encode graph topology and relation type inside attention and message-passing systems (Ying et al., 2021; Hu et al., 2020; Rampášek et al., 2022). Knowledge Sheaves and Neural Sheaf Diffusion establish local vector spaces, restriction maps, and consistency-oriented computation (Gebhart et al., 2021; Bodnar et al., 2022). These works own the general idea that relational organization can be a useful inductive bias; P9 does not introduce it.

Algorithmic and hybrid computation. CLRS provides a broad benchmark for neural algorithmic reasoning, while TransNAR combines a language Transformer with a graph neural algorithmic reasoner (Veličković et al., 2022; Bounsi et al., 2024). Compositional Choke Points goes further in the direction relevant here by using a deterministic benchmark and symbolic–neural oracle analysis to isolate failures of iterative computation (Anonymous, 2026). Our explicit-inference arm should therefore be read as a bounded diagnostic control, not a new symbolic fallback architecture.

Reusable mechanics, latent reasoning, and mechanism models. DreamCoder and LILO learn reusable program abstractions and libraries (Ellis et al., 2020; Grand et al., 2023); Neural Production Systems factor rule templates from entity bindings (Goyal et al., 2021); Coconut studies continuous latent reasoning (Hao et al., 2024); and mechanistic world-model work argues for reusable explanatory mechanisms in scientific representations (Posner et al., 2026). These are donor families, not novelty targets for the present paper. Our final study does not promote a learned module library, dedicated binding mechanism, recurrent latent reasoner, or causal-discovery architecture.

Representation diagnosis, missing information, and serialization. Temporal neuro-symbolic QA has been used to distinguish representation inconsistency from reasoning failure (Liem, 2026), while Quest-

Bench tests whether a model can ask for missing information rather than answer an underspecified query (Li et al., 2025). Most directly for our D1 control, serialization friction shows that preserving underlying structured content while changing its serialization can materially change performance (Lo et al., 2026). We therefore do not claim the general principle that native relational structure can outperform a flat serialization. D1 instead asks whether that distinction remains load-bearing on our exact method-coordinate transfer protocol, with the semantic fields matched across the two typed arms.

3 Problem formulation

3.1 Structured worlds and restricted views

A benchmark world contains typed atoms, typed relations, optional local affine representation maps, candidate mechanics, admitted failure history, and an evaluator-only target. The learner never receives the protected target or protected generation metadata. Five progressively richer views are exposed:

$$V_{\text{surface}} \subset V_{\text{topology}} \subset V_{\text{typed}} \subset V_{\text{current}} \subset V_{\text{semantic}}.$$

Surface contains reminted surface structure; topology retains connectivity without semantic relation labels; typed adds semantic atom/relation/mechanic coordinates; current additionally exposes local transport values; semantic also admits scoped negative history.

For a finite protected sample $\mathcal{D}$ and view V, let $f_V(x)$ be the exact model-visible fingerprint. Partition $\mathcal{D}$ into groups g of equal fingerprints. The empirical deterministic ceiling is

$$C_V = \frac{1}{|\mathcal{D}|} \sum_g \max_y |\{x \in g : y(x) = y\}|.$$

This is elementary accounting over the frozen equivalence classes. It is not a new theorem about Bayes error, sufficient statistics, state aliasing, or asymptotic learnability. Its role is operational: a model above this ceiling indicates leakage or an evaluator defect; a model at the ceiling of a non-identifying view cannot be improved deterministically without changing the information supplied or the task.

3.2 Candidate-relative mechanic selection

Mechanic identities are reminted across worlds and splits. The learned task therefore scores a current candidate m_i relative to context E,

$$s(E, m_i) \in \mathbb{R},$$

rather than classifying against a global mechanic vocabulary. Candidate presentation order is derived from declared model-visible information and an independent seed. This rule was frozen after a pre-outcome hostile review found that ordering candidates using a richer hidden fingerprint could leak semantics through presentation position.

3.3 Local affine gluing

The D0 transport family uses exact one-dimensional affine maps between local charts. A complete directed cycle composes to

$$x \mapsto ax + b.$$

The protected target is GLUE when the composition is identity under the frozen exact/tolerance rule, OBSTRUCTION when a complete unique cycle is non-identity, and UNKNOWN when visible information does not determine a unique complete composition. The family is intentionally small. Its purpose is to create a transparent global-composition discriminator, not to claim general sheaf-theoretic reasoning.

4 Experimental design

4.1 D0 hostile information pairs

Three paired families isolate different load-bearing coordinates while holding weaker views fixed.

Relation semantics. Two worlds share weaker structure but differ in whether an evidence relation supports or defeats a claim, changing which candidate effect is appropriate.

Local transport. Two worlds share typed endpoints and topology but differ in affine map values, yielding GLUE versus OBSTRUCTION. Source-insufficient cases remain UNKNOWN.

Admitted failure history. Two worlds share the current typed state and candidate set but differ in scoped admitted negative history. Only the semantic view contains the decisive historical coordinate.

Hand-authored fixtures were generalized into deterministic train/dev/test corpora with opaque content-derived identities, split-disjoint namespaces, evaluator/model metadata separation, and exact manifests. These data are exact traces, not pseudo-gold extracted from natural scientific papers.

4.2 M0 evaluator hardening

Before learned execution, M0 froze a shared task interface, null predictors, an exact semantic oracle, and leakage sentinels. Pre-outcome reviews corrected two defects that are retained in the audit history: hidden semantic state initially influenced candidate order, and an early task adapter removed incidence information promised by the surface view. The result-bearing protocols use the corrected interfaces. The oracle is tested against falsified evaluator gold and is a harness ceiling, not a competitive model.

4.3 M1: simple-model first right of refusal

M1 uses generic classical models before any neural escalation: logistic regression, decision trees, random forests, and nearest neighbours. Mechanic selection uses field features and context–candidate interactions without a hard-coded global candidate vocabulary. Gluing uses generic field features and, when map values are visible, a canonical sequence of local affine coefficients; it does not compute the composed residual directly. Hyperparameters are selected using train → dev only, and each selected configuration receives one protected test evaluation. Candidate-order and row-order invariance, label-shuffle checks, and per-task information ceilings are mandatory.

Two implementation amendments occurred before protected M1 outcome access and did not alter the scientific protocol: a runtime compatibility repair made ‘DictVectorizer‘ output dense arrays after scikit-learn 1.9 rejected CSR index metadata, and a later smoke-test-only repair increased the tiny smoke fixture so the already-frozen $k = 15$ nearest-neighbour configuration was defined. The official 64/24/48 train/dev/test corpus sizes, feature semantics, model grid, dev selection, protected test, and terminal rules were unchanged.

4.4 Explicit inference first right of refusal

When a residual is algorithmically transparent, a separately frozen payload-only procedure receives first right of refusal before richer learned architectures are considered. The A5 transport procedure reconstructs the visible local cycle and composes unique affine maps exactly. A2 maps visible typed relation semantics to declared candidate effects. A4 filters candidates using only admitted scoped failure history, with an explicit fallback when no such history is visible. These controls do not establish that symbolic inference is generally superior; they test whether neural complexity is necessary for these bounded atoms.

4.5 D1: whole-domain procedural transfer

D1 moves from generated state worlds to exact authored method structures. Training domains are numerical methods and graph algorithms; the entire protected test domain is transactional workflows. Surface

Table 1: M1 protected accuracy versus the exact empirical deterministic ceiling induced by each frozen view.

View	Accuracy	Ceiling	Gap
SURFACE	0.500000	0.500000	0.000000
TOPOLOGY	0.500000	0.500000	0.000000
TYPED	0.666667	0.666667	0.000000
CURRENT	0.670139	0.833333	0.163194
SEMANTIC	0.836806	1.000000	0.163194

action identities are reminted and domain-disjoint. Training/dev contain aligned examples, single-coordinate obstructions, and source-insufficient UNRESOLVED examples. The protected test additionally includes multi-coordinate corruptions whose constituent coordinates appeared separately in training.

Four arms receive progressively different organization of the data:

1. reminted transcript/action bag;
2. untyped pair/topology features;
3. explicit typed relational comparison features;
4. a same-information typed canonical serialization without the explicit relational comparison.

The last contrast is central. It does not test whether the serialized arm lacks the semantic fields; it tests whether the relational organization of those matched fields is load-bearing on this protocol. Because serialization friction is established prior work (Lo et al., 2026), the result is interpreted only at this application-specific scope.

The D1 protocol was amended before results to compare dependencies by method-local role topology rather than literal action names and to make dependency corruptions reverse an actually present edge rather than silently no-op.

4.6 Verification and authority boundaries

Each result-bearing study had a pre-artifact independent expectation frozen before official outcome access. Official results were compared to those expectations and preserved with execution and verification receipts. A2/A4 and D1 have zero discrepancies. M1 has one retained non-material discrepancy: the independent implementation expected a random-forest-class selected model, while the official selected nearest-neighbour model tied the random forest exactly on dev accuracy and won under the pre-frozen lower-complexity tie-break. Material discrepancy count is zero.

All outputs are diagnostic. No model score can grant scientific truth, novelty, execution permission, or adoption authority.

5 Results

5.1 M1 separates information from computation

Across the frozen view ladder, M1 never exceeds its empirical deterministic ceiling. Surface and topology both attain their ceilings (0.500000/0.500000 and 0.500000/0.500000, respectively), while typed reaches 0.666667 at a ceiling of 0.666667. Richer views expose more information but also reveal a residual: current obtains 0.670139 against an overall ceiling of 0.833333, and semantic obtains 0.836806 against 1.000000.

The task decomposition identifies the source of that gap. Semantic mechanic ranking reaches 1.000000, so the visible typed relation/history coordinates are sufficient and exploited on the mechanic task. In contrast, semantic affine gluing remains at 0.510417 despite an exact task ceiling of 1.000000; current gluing is likewise

Table 2: D1 protected whole-domain transfer to transactional workflows. The typed serialization exposes the same typed semantic fields as the relational arm but removes the explicit relational comparison. Exact selected-model identifiers remain in the evidence archive.

Arm	Acc.	Macro-F1	Double corrupt.	UNRESOLVED
Transcript bag	0.250000	0.133333	0.000000	0.000000
Untyped pair	0.906250	0.912886	1.000000	1.000000
Typed relational	1.000000	1.000000	1.000000	1.000000
Typed serialization	0.500000	0.222222	1.000000	0.000000

Table 3: Post-hoc paired uncertainty analysis of the frozen 128 protected D1 predictions. This quantifies the existing endpoint; it is not a new preregistered endpoint.

Comparator	Delta	Paired 95% CI	Wins–losses	McNemar p
Transcript bag	0.750000	[0.671875, 0.820312]	96–0	2.524×10^{-29}
Untyped pair	0.093750	[0.046875, 0.148438]	12–0	0.000488
Same-information typed serialization	0.500000	[0.414062, 0.585938]	64–0	1.084×10^{-19}

0.510417. This is the pattern expected from a computation residual rather than an information deficit on the frozen benchmark. The general idea of separating representation from downstream reasoning failure has close parents (Liem, 2026; Anonymous, 2026); our claim is limited to this prospectively controlled instance.

5.2 Explicit computation closes the D0 residual

A5 confirms that the residual need not motivate a richer neural architecture. With only typed endpoints and no local map values, the payload-only affine procedure remains unresolved (full-task accuracy 0.000000). Once the current view exposes the local affine maps, exact cycle composition reaches 1.000000. Thus the operation missed by M1 is exactly computable from visible coordinates in this D0 family.

A2 and A4 show a complementary pattern. Topology alone gives 0.000000 on the relation hostile pair, while typed relation semantics yield 1.000000 using the explicit relation selector. On the history family, current-state information gives 0.500000, while the semantic view with admitted scoped failure history yields 1.000000. These results establish that the coordinates are load-bearing on their exact hostile pairs, not that neural mechanic or memory modules are required.

5.3 Typed relational structure transfers across a whole held-out domain

D1 supplies the external procedural-transfer test. The typed relational arm reaches 1.000000 protected accuracy on transactional workflows, compared with 0.906250 for untyped pair structure, 0.500000 for the same-information typed serialization, and 0.250000 for the reminted transcript bag. The improvement over the same-information serialization is 0.500000. The 0.750000 figure against transcript is reported separately and is not a transfer margin: the transcript comparator answers ALIGNED on all 128 protected cases, so the quantity is $1-\text{prior}(\text{ALIGNED})$. Its content is that a surface-action view has an empty transferable vocabulary over a whole-domain holdout, which is a stronger and narrower statement than outperforming it. The typed relational arm also preserves 1.000000 accuracy on double corruptions and 1.000000 on source-insufficient UNRESOLVED cases.

The same-information contrast was originally the important one for interpretation, and a hostile check has since withdrawn it. The serialized arm does contain the same typed semantic fields — all 512 of its token lists decode back to their typed payload exactly, which we have now verified rather than assumed — so the 0.500000 difference cannot be described as simple information addition. But it also cannot be described as explicit relational comparison making those fields more useful, because the serialized arm's protected answers are a function of the symbol alphabet rather than of the semantics. Applying a bijection to the value alphabet — 220 atoms to 220 distinct images, every gold label preserved — moves that arm from 0.75

to 0.50 accuracy and from 0.5 to 0.0 informedness, changing 32 of 128 protected predictions, while renaming nothing but its feature keys. Its published figure is one point of that orbit. Carrying the information is not the same as answering from it, and this comparator does not answer from it; the difference therefore measures which symbols were drawn. The surviving relational margin is the typed-versus-untyped contrast, whose two arms are invariant under the same transform. Because the general fact that serialization can affect reasoning is established prior work (Lo et al., 2026), we do not claim a universal representation theorem from D1.

As a post-hoc robustness analysis of the already-frozen D1 endpoint, we compared the selected typed relational arm to each comparator on the identical 128 protected cases. Against transcript, the accuracy difference is 0.750000 with paired bootstrap 95% interval [0.671875, 0.820312]; all 96 discordant cases favor typed relational and none favor transcript (exact McNemar $p = 2.524 \times 10^{-29}$). Because the transcript arm is constant on these cases, that comparison is against a majority-class predictor and should be read as such rather than as a contrast between two fitted views. Against the same-information typed serialization, the difference is 0.500000 with paired interval [0.414062, 0.585938] and discordance 64–0 ($p = 1.084 \times 10^{-19}$). Against the untyped pair arm, the paired interval is [0.046875, 0.148438] with discordance 12–0 ($p = 4.883 \times 10^{-4}$). These intervals and exact tests were not preregistered primary endpoints; they quantify the frozen case-level predictions without changing the terminal.

6 What the result does and does not establish

The evidence supports a bounded staged diagnosis:

1. some weak views are non-identifying on exact hostile pairs;
2. typed relation and admitted-history coordinates remove those particular collisions;
3. simple learned models can exploit some of the exposed structure, especially candidate mechanic choice;
4. one remaining D0 residual is global affine composition and closes under a frozen exact procedure;
5. typed relational method coordinates transfer to a held-out procedural domain better than untyped and same-information serialized comparators on D1, and the reminted transcript view carries no transferable vocabulary over that holdout.

The evidence does *not* establish a universal sufficient representation for reasoning, a new neural architecture, a general graph or sheaf learner, a causal-discovery system, a reusable tactic library, or a proof that simple models dominate neural systems. The whole-domain transfer set contains only 128 protected cases and three method families. The exact worlds are intentionally synthetic and small. D1 broadens the domain boundary but remains exact, authored method structure rather than open-world scientific text.

One axis of the hostile analysis was never testable, and the reason is a property of the data model rather than of the arms. A third hostile variant reverses every sequence-valued comparison coordinate and was intended to ask whether any arm keys on presentation order within a coordinate. It cannot: the method-realization builder normalises each of those coordinates to `tuple(sorted(set(...)))`, so the reversal is undone by the rebuild and the variant's dataset is identical to the base, manifest digest included. Ordering — and, by the same normaliser, multiplicity — is not representable in the comparison coordinates at all. The eight feature families are consequently order-blind by construction rather than by choice, and no result here should be read as evidence that they resist reordering; no ordering was ever presented to them. The bijection and equal-length variants do reach their arms, on 512 and 112 of 512 instances respectively, so this is a statement about one variant and not about the analysis. Measuring order sensitivity would require the comparison coordinates to carry an order first.

The model-escalation conclusion is procedural, not architectural: the frozen evidence never produces a residual that requires a richer learned model after information and explicit-computation controls. Consequently the terminal is `P9_NEURAL_ESCALATION_NOT_JUSTIFIED`, not `NEURAL_MODELS_FAIL`.

7 Reproducibility

The repository contains the exact generated corpora, protocols, run scripts, selected configurations, protected predictions, result digests, independent expectations, comparison receipts, claim ledgers, and the TMLR package. A fail-closed summary builder refuses to produce publication metrics if any required official result, terminal, digest, verification receipt, hostile check, or current integration-authority receipt is absent. Tables and manuscript macros are generated only from that summary. From a clean checkout, the reproduction path is:

```
python papers/paper-09-structured-epistemic-learning/reproduce_final.py
```

The script regenerates the post-hoc paired D1 analysis, independently compares the integrated official result archives to the frozen expectation table, rebuilds the fail-closed evidence summary, result macros and headline tables, and runs the manuscript audit. Runtime dependencies are pinned in the repository environment; the raw M1 and D1 protected archives are committed rather than reconstructed from manuscript numbers.

This paper is intentionally conservative about authority. Execution receipts prove that named computations occurred and matched bounded expectations. They do not grant scientific truth or adoption authority. Historical donor receipts that referred to superseded pull requests remain provenance, while the current integration receipt separately authorizes rendering of the exact grafted artifacts on this review branch.

8 Conclusion

P9 asks whether richer learning machinery is needed after controlling what information a model can see. On exact hostile pairs, weaker views impose observable collisions; typed relation and admitted-history coordinates remove the targeted ambiguities. On M1, generic classical learning then succeeds on mechanic selection but leaves a global affine-composition residual. A separately frozen exact procedure closes that residual from visible local maps. Finally, typed relational method coordinates transfer across a whole held-out procedural domain and outperform the untyped and same-information serialized controls, while the reminted transcript control has no transferable vocabulary to outperform. The resulting conclusion is narrower than an architecture proposal but stronger than a benchmark score: before escalating model complexity, identify whether the failure is informational, statistical, computational, or transfer-related.

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